HostelCare Run and CRUD Demo Guide

All servers start karne, site/app open karne, ani Postman madhun JSON Server + Spring Boot CRUD live dakhavnyasathi.

Soppa niyam: pratyek service veglya VS Code terminal madhye chalu kara ani demo sampestovar terminals band karu naka.

1. Konti service kashasathi?

Service	Kaam	Address / port
JSON Server	HostelCare complaints; Postman, site ani React madhye shared data.	http://localhost:3000
Spring Boot	Residents CRUD; complaints routes JSON Server kade forward karto.	http://localhost:8080
Flask JWT	Register/login, JWT bearer token, protected dashboard.	http://127.0.0.1:5000
Vite	Original website ani React portal eka origin var serve karto.	http://localhost:5173

2. Sagla punha chalu kara - ya kramane

VS Code madhye Hostelcare folder open kara. Terminal > New Terminal vaprun char terminals ughada. Pratyek terminal madhye khalcha command group vegvegla chalva.

Terminal 1 - JSON Server (complaints)

```
cd "D:\my learning\Hostelcare\exp2-api"
npm start
```

Terminal 2 - Spring Boot (residents)

```
cd "D:\my learning\Hostelcare\exp7-spring-api"
& "D:\download\apache-maven-3.10.0-bin\apache-maven-3.10.0\bin\mvn.cmd" spring-boot:run
```

Terminal 3 - Flask JWT login API

```
cd "D:\my learning\Hostelcare\exp8-jwt"
.\.venv\Scripts\python.exe app.py
```

Terminal 4 - Website + React portal

```
cd "D:\my learning\Hostelcare\exp3-react"
npm run dev
```

JSON Server aadhi chalu kara. Pratyek terminal madhye server ready/run message disla pahije. Service vapartana terminal chalu theva.

3. Browser madhye site ani app ughada

Vite terminal chalu zalyavar donhi link port 5173 var ughada. Ekyach origin mule browser madhil HostelCare account storage website ani React portal madhye share hote. Live Server cha port 5500 vaparla tar storage vegla hoil.

```
Website:     http://localhost:5173/
Student login: http://localhost:5173/user-login.html
Admin login:   http://localhost:5173/admin-login.html
React portal:  http://localhost:5173/exp3-react/
```

Server chalu aahet ka te baghnyasathi

Browser address	Expected result
http://localhost:3000/complaints	JSON Server complaints list
http://localhost:8080/api/students	Spring Boot residents list
http://localhost:8080/api/health	Spring Boot health status
http://127.0.0.1:5000/health	Flask JWT status: ok

4. Postman - Experiment 2 complaints CRUD

Postman madhye request method nivda, URL taka, ani Send daba. Write requests sathi Body > raw > JSON nivda. Khali dilela ID fakt udaharan aahe; POST response madhun milalela ID pudhchya requests madhye vapra.

Operation	Method and URL	Expected
List all	GET http://localhost:3000/complaints	200 OK
Get one	GET http://localhost:3000/complaints/1	200 OK
Create	POST http://localhost:3000/complaints	201 Created
Replace full record	PUT http://localhost:3000/complaints/{id}	200 OK
Change selected fields	PATCH http://localhost:3000/complaints/{id}	200 OK
Delete	DELETE http://localhost:3000/complaints/{id}	200 OK

POST body - navi complaint tayar hote:

```
{
  "studentName": "Aarav Deshmukh",
  "studentEmail": "aarav.test@example.com",
  "roomNumber": "A-204",
  "category": "Electrical",
  "priority": "Medium",
  "description": "Ceiling fan is making noise.",
  "date": "2026-10-22",
  "status": "Pending"
}
```

5. Postman - JSON Server PUT / PATCH / DELETE

POST madhun milalela complaint ID {id} chya jagi taka. PUT sampurna record replace karto; sagle fields dya. PATCH fakt pathavlele field badalto.

PUT body - sampurna complaint record pathva:

```
{
  "studentName": "Aarav Deshmukh",
  "studentEmail": "aarav.test@example.com",
  "roomNumber": "A-204",
  "category": "Electrical",
  "priority": "High",
  "description": "Fan is making noise.",
  "date": "2026-10-22",
  "status": "In Progress"
}
```

PATCH body - fakt badal karaychi field:

```
{"status":"Resolved"}
```

PATCH nantar GET kara ani badal paha. DELETE nantar tya ID cha GET 404 dakhavto ka check kara. Site ani React dashboard sadharan 2 seconds madhye update hotat.

6. Postman - Spring Boot resident CRUD (Experiment 7)

Operation	Method and URL	Expected
List residents	GET http://localhost:8080/api/students	200 OK
Create resident	POST http://localhost:8080/api/students	201 Created
Get one resident	GET http://localhost:8080/api/students/{id}	200 OK
Replace resident	PUT http://localhost:8080/api/students/{id}	200 OK
Change selected fields	PATCH http://localhost:8080/api/students/{id}	200 OK
Delete resident	DELETE http://localhost:8080/api/students/{id}	200 OK

POST ani PUT sathi sample JSON:

```
{
  "name": "Aarav Deshmukh",
  "email": "aarav@example.com",
  "course": "Computer Engineering",
  "roomNumber": "A-204",
  "mobile": "9876543210"
}
```

PATCH example: {"roomNumber":"A-205"}. POST response madhil ID pudhchya GET/PUT/PATCH/DELETE sathi vapra. Postman madhle resident changes Spring Boot chaloo astana admin residents list madhye disatil.

7. Spring Boot complaints endpoints

Spring Boot complaint routes pan aahet; te JSON Server la forward kartat. Don API terminals - port 3000 ani 8080 - donhi chalu havet.

Method	URL
GET	http://localhost:8080/api/complaints
GET one	http://localhost:8080/api/complaints/{id}
POST / PUT / PATCH / DELETE	http://localhost:8080/api/complaints (ID pathavtana /{id})

8. Postman - Experiment 8 JWT test

Collection missing asel tar Postman madhye Import kara ani ha file nivda:

```
D:\my learning\Hostelcare\exp8-jwt\HostelCare-Exp8-JWT.postman_collection.json
```

Collection madhye requests ya kramane Send kara:

Kram	Request	Expected status
1	Register Student	201 Created
2	Login Student and Save JWT	200 OK; token save hoto
3	Protected Dashboard	200 OK
4	Dashboard Without Token	401 Unauthorized
5	Dashboard With Tampered Token	401 Unauthorized

Flask server restart zalyavar saved JWT vapru naka. Login request punha run kara. Flask account sapadla nahi tar Register Student punha run kara; duplicate email asel tar navin email nivda.

9. Data ani stop/start lakshat theva

Data / service	Band kelyavar kay hota?
Complaints - exp2-api/db.json	Records file madhye save hotat; server restart nantar tikun rahtat.
Spring Boot residents - H2 memory DB	Spring Boot restart nantar direct Postman records reset hotat. Browser accounts available astil tar site sync karu shakate.
Flask JWT accounts/token	Flask restart nantar API accounts/tokens reset hotat. Site session close zalyavar login punha kara; jar credentials invalid astil tar register punha kara.

Sarva services band karayla tyanchya pratyek terminal madhye Ctrl + C daba. Maven terminal-la "Terminate batch job (Y/N)?" vicharle tar Y daba. JSON Server complaints db.json madhye save hotat.